AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM129
Technologies in practice

Midterm Examination (MTA)
Fall Semester 2025/2026

Date:17/Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 H

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Instructions:

- Total Marks:**60**
- This exam consists of **4** Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [20 marks]: Choose The correct answer

1. **What is the primary purpose of data transmission in networking?**
   a) To store data permanently on devices
   b) To convert digital signals into analog for long-distance travel
   c) To encrypt data before sending it over cables
   d) To compress files for faster upload
2. **In the encapsulation process, what is added to the data at the Transport Layer?**
   a) Physical addressing (MAC)
   b) Segment header with port numbers
   c) Frame trailer with CRC
   d) Packet header with IP addresses
3. **What is the role of the Ethernet frame's CRC field?**
   a) To identify the network type
   b) To detect errors in the transmitted data
   c) To route the frame to the correct VLAN
   d) To compress the payload
4. **Which protocol is commonly used for dynamic IP addressing in IPv4 networks?**
   a) ARP
   b) DHCP
   c) ICMP
   d) SNMP
5. **What is a key advantage of IPv6 over IPv4 addressing?**
   a) Shorter address length for easier memorization
   b) Built-in support for NAT to conserve addresses
   c) Larger address space using 128 bits
   d) Reliance on broadcast for device discovery
6. **In a typical home network, what device connects the LAN to the ISP?**
   a) Switch
   b) Modem
   c) Access Point
   d) Repeater
6. **Which topology is most commonly used in modern home Wi-Fi networks for better fault tolerance?**
   a) Bus
   b) Ring
   c) Star
   d) Mesh (full)
8. **What is the main function of the operating system kernel?**
   a) Providing a graphical user interface
   b) Managing hardware resources and acting as intermediary between apps and hardware
   c) Compiling user programs
   d) Encrypting files for storage
9. **Which OS operation ensures isolation between multiple virtual machines?**
   a) Multitasking
   b) Virtualization
   c) Paging
   d) Deadlock prevention

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10. **What is the purpose of a system call in an operating system?**
    a) To debug application code
    b) To request kernel services like file I/O from user programs
    c) To load dynamic libraries at compile time
    d) To boot the OS from ROM

Question 2: Compare between the following [10 marks]:

1. Static addressing vs. Dynamic addressing in IP networks (2 Marks)
2. User-level threads vs. Kernel-level threads in operating systems (2 Marks)
3. Encapsulation vs. Decapsulation in network data transmission (2 Marks)
4. Batch processing vs. Time-sharing in operating systems (2 Marks)
5. 2.4 GHz vs. 5 GHz Wi-Fi bands in home networks (2 Marks)

Question 3 [10 marks]

1. Match the networking protocols to their primary functions.(4 Marks)

Protocol	Function
1. FTP	a) Transfers files over a network
2. HTTP	b) Sends emails across networks
3. SMTP	c) Delivers web content to browsers
4. SNMP	d) Manages network devices remotely

2. What is the role of a process scheduler in an operating system? (3 Marks)
3. What is the purpose of a file allocation table (FAT) in a file system? Name two file systems that use it (3 Marks)

Question 4 [10 marks]

1. Match the networking protocols to their primary functions. (4 Marks)

Protocol	Function
ICMP	a) Monitors network device performance
OSPF	b) Diagnoses network connectivity issues (e.g., ping)
BGP	c) Routes data within an autonomous system using shortest paths
SNMP	d) Exchanges routing info between different networks on the internet

2. What is the purpose of system calls in operating systems? Provide two examples. (3 Marks)
3. What is resource management in OS services? Explain with an example. (3 Marks)

Question 5 [10 marks]

1. What is the purpose of a linker and a loader in an OS?(3 Marks)
2. What is resource allocation in OS services? (3 Marks)
3. Contrast the security capabilities of WPA2 and WEP protocols in securing wireless networks. What are the advantages and limitations of each?(4 Marks)

Question 6 [10 marks]

1. What is operating system debugging, and why is it essential?(4 Marks)
2. Why is NAT (Network Address Translation) used in IPv4 networks? (3 Marks)
3. Explain in details Virtualization in modern computing environments ?(3 Marks)

End of Questions

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